The Mathematics of Recurrent Neural Networks

This reading provides you with the opportunity to check your understanding of the mathematics of recurrent neural networks (RNNs). We will provide you with network parameters and an input and suggest you try to compute the output by hand first, before looking at the correct solution on the next page. Enjoy!

Let's consider the following one-word sentence: *hello* For processing, we add start-of-sequence and end-of-sequence tokens, and our goal is to encode the sentence using an RNN.

Given Parameters and Input

The embeddings corresponding to our tokens are:

$$\text{hello} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \quad \langle s \rangle = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \quad \langle s \rangle = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

The initial hidden state is:

$$h_0 = \begin{bmatrix} 1 \end{bmatrix}$$

The weight matrices of the RNN are:

$$W_x = \begin{bmatrix} 1 & 5 & 3 \end{bmatrix}, \quad W_h = \begin{bmatrix} 7 \end{bmatrix}$$

We assume an RNN without any bias terms.
Compute the RNN's hidden state corresponding to $\langle s \rangle$! (Reminder: We suggest you try this on your own before checking the solution on the next page.)

Solution: Encoding of $\langle s \rangle$

The hidden state is computed as:

$$
\begin{aligned}
h_1 &= \text{ReLU}(W_h h_0 + W_x \text{emb}(\langle s \rangle)) \\
&= \text{ReLU}\left(\begin{bmatrix} 7 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} + \begin{bmatrix} 1 & 5 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right) \\
&= \text{ReLU}\left(\begin{bmatrix} 7 \end{bmatrix} + \begin{bmatrix} 16 \end{bmatrix} \right) \\
&= \text{ReLU}\left(\begin{bmatrix} 23 \end{bmatrix} \right) \\
&= \begin{bmatrix} 23 \end{bmatrix}
\end{aligned}
$$